Baochen Tian

Research Assistant, Australian National University
baochen.tian.phys@gmail.com | github.com/bcTiann | Website: bctiann.github.io

EDUCATION

Australian National University Graduated 2025

Bachelor of Science in Physics, 2023 - 2025 | GPA: 6.13 / 7.0
Exchange: National Taiwan University, Spring 2025

Beijing Institute of Technology 2020 - 2023

Undergraduate studies; transferred to ANU in 2023
International Economics and Trade (2020 - 2021); Optical-Electronics Information Science (2021 - 2023)

PERSONAL PROJECTS

SuperMimic Reproduction October 2026

Trained and evaluated the differentiable world model and policy in SuperMimic, based on UniSuperMimic, on 40 LAFAN1/G1 motions in MuJoCo.
11,000-update checkpoint: 40/40 full clips completed; macro normalized return 0.826 (one seed; training motions).
Code: github.com/bcTiann/Uni-SuperMimic | Results

GeoRA Implementation and GRPO Experiments October 2026

Implemented GeoRA from the paper and integrated it with GRPO training of Qwen2.5-1.5B-Instruct; compared against LoRA at the same trainable-parameter budget.
After 25 updates, MATH500 accuracy was 56.8% versus 53.8% for LoRA (+3.0 percentage points; single seed).
Code: github.com/bcTiann/geora-repro | Results

Flow Matching Reimplementation of DiffPhyCon 2026

Derived and implemented a flow-matching formulation of the joint trajectory model in DiffPhyCon (NeurIPS 2024), a diffusion-based physical control framework. Matched original model capacity and training configuration; compared 1D Burgers control and 2D Jellyfish state-control trajectories with diffusion baselines.
Investigated guidance and prior reweighting under flow matching using score-to-velocity conversions.
Code: github.com/bcTiann/diffphycon

RESEARCH EXPERIENCE

Python Pipeline for Post-processing QUOKKA Simulations July 2025 - Present

Research School of Astronomy and Astrophysics, ANU.
Research Assistant (April 2026 - Present). Supervisor: Dr. ChongChong He.

Developing a reusable Python post-processing pipeline for QUOKKA, a next-generation GPU-native astrophysical hydrodynamics code. Covers 3D data loading, chunked computation, analysis and visualisation; produces synthetic images, spectra and statistics for comparisons with astronomical observations and physical model tests.
Using DESPOTIC, Cloudy and analytic line-emission models, with thermal broadening and dust attenuation, to investigate how reliably CO, H-alpha, C+ and H I line widths recover mass-weighted gas velocity dispersion.
Code: github.com/bcTiann/quokka_postprocess_3D

Deep Learning for Cosmic Microwave Background Inpainting February - June 2025

National Taiwan University (exchange). Supervisor: Prof. Pisin Chen; mentor: Reyhan D. Lambaga.

Applied Transformers and DDPMs to inpaint masked cosmic microwave background maps; evaluated reconstruction through angular power spectra against ground truth.

Axion Haloscope Modelling (EQUS Scholarship) January - February 2024

University of Western Australia. Supervisors: Prof. Michael Tobar and Dr. Aaron Quiskamp.

Six-week scholarship project: COMSOL modelling of resonant modes and sensitivity for tunable-wall rectangular haloscopes.

SKILLS & INTERESTS

Deep learning: PyTorch, Transformers, diffusion models, flow matching, GRPO and LoRA/GeoRA.
Scientific computing: Python, MuJoCo, QUOKKA post-processing, DESPOTIC and Cloudy.
Chamber music: Co-organised ensembles with friends at BIT and ANU; choir accompanist and public performances.